

# Carbon Mass Balance Evaluation of Cellulase Production on Soluble and Insoluble Substrates

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A methodology is described and applied for performing carbon mass balances across cellulase enzyme production processes using both soluble sugar and insoluble cellulose substrates. The fungus *Trichoderma reesei* was grown on either glucose, lactose, or cellulose in aerobic batch mode, and the evolution of the main carbonaceous components (cell mass, cellulose, soluble protein, adsorbed protein, sugars, and carbon dioxide) was followed. A variety of analytical techniques were utilized to measure these components, including (i) gravimetric analysis, (ii) near-infrared spectroscopy, (iii) bicinchoninic acid based soluble protein measurement, (iv) gas mass spectrometry and flow rate, (v) CHNS/O elemental analyses, and (vi) high-performance liquid chromatography. The combined set of measurements allowed carbon mass balances across the cellulase production process to be assessed to determine the consistency of the underlying kinetic data. Results demonstrate the capability to determine the levels and distribution of all major carbonaceous components during the cellulase production process on both soluble and insoluble substrates. Average carbon mass balance closures were near 100% during early stages (<72 h) of the cultivations using glucose, lactose, or cellulose as the substrates, but carbon mass closures trended high later in the cultivation. Analysis of carbon allocation results suggests that an error in the gas mass flow rate measurement was the primary cause for carbon mass balance closures to exceed 110% late in the process.

## Introduction

The importance of an efficient and cost-effective ethanol from biomass (bioethanol) production process technology has been acknowledged for a long time (Wooley et al., 1999). Established technologies for bioethanol production are crop-based, utilizing substrates such as sugar cane juice and cornstarch. Since the cost of raw materials can be as high as 40% of the production cost (Zaldivar et al., 2001), recent efforts are concentrated on using less expensive lignocellulosic materials (e.g., forestry and agricultural residues, grasses, etc.) containing substantial amounts of the carbohydrate polymers cellulose and hemicellulose. The abundance of these biopolymers in nature makes lignocellulosic materials important resources for the large-scale production of bioethanol.

Cellulase enzymes hold promise for use in bioethanol production processes based on enzymatic cellulose saccharification because of their potential to achieve high glucose (cellulose) conversion yields. Unfortunately, commercial cellulase enzymes are still too expensive to be used for the conversion of waste lignocellulosics to liquid fuels, especially ethanol (Hayward et al., 2000). Installed

equipment costs for on-site cellulase enzyme production by themselves are estimated to represent 11% of the overall capital costs of a large-scale enzyme-based bioethanol production process (Wooley et al., 1999). While current cellulase costs are high, however, it is believed that production costs can be significantly reduced by applying the tools of modern biotechnology and process engineering (Hayward et al., 1999; Kansoh et al., 1999; Tholudur et al., 1999; Nikolov et al., 2000; Schell et al., 2001).

To achieve economic threshold performance values for cellulase production requires advancing our understanding of the production process. To this end, high quality performance data validated by accurate carbon mass balance closures is desired. Figure 1 shows a generic representation of cellulase production by a fungal biocatalyst utilizing insoluble cellulose as the primary carbon source. As this figure shows, calculating carbon mass balances during the aerobic protein production process requires knowing the amount of carbon contained in all carbonaceous substrates and products, namely, fungal cell mass, cellulose, soluble protein, adsorbed protein, soluble sugar (cellobiose, glucose), media components, and carbon dioxide (CO<sub>2</sub>).

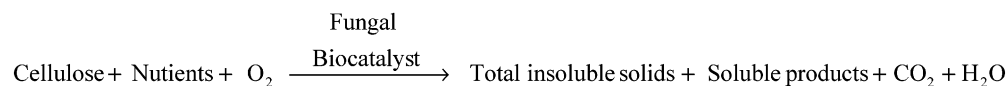
Carbon mass balancing a biobased production process is a critical and necessary technique for assessing the accuracy of performance measurements (e.g., product yields). Obtaining accurate carbon mass balance closure

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Total insoluble solids = fungal cell mass (mycelium and spores) + residual cellulose + adsorbed protein + insoluble nutrient medium components

Soluble Products = residual sugars + soluble protein + byproducts

**Figure 1.** Schematic reaction of cellulase enzyme production using a fungal biocatalyst growing on cellulose.

provides confidence (but does not conclusively prove) that all of the important carbonaceous components in the system are being properly accounted for. Thus, achieving good carbon mass balance closures can be used to justify the worthiness of developing predictive mathematical models of the fermentation process based on these data. The technique of carbon mass balancing has been found beneficial in two primary ways. First, as illustrated in this paper, carbon balancing helps elucidate and troubleshoot potential problems in experimental systems designed to measure the performance of complex conversion processes. Second, verifying internal data consistency is a prerequisite to developing and validating accurate morphologically structured models of industrially important fermentation processes (Blanch and Clark, 1996) such as those for the aerobic production of commercial antibiotics (e.g., ampicillin, bacitracin, amoxicillin, etc.) and enzymes (e.g., proteases, amylases, cellulases, etc.).

Despite wide historical use, however, there are few reports of carbon balances (C-balances) on cellulase enzyme production processes, particularly those employing insoluble substrates. Ross et al. (1983) reported C-balances for a two-stage continuous cultivation of *Trichoderma reesei* (T. reesei) QM9414 using purified cellulose (Avicel) as the primary carbon source substrate. Their C-balance accounted for substrate, cell mass, enzyme, and CO<sub>2</sub> production, and they reported C-balance closures ranging from 60% to 110% and from 120% to 130% (estimated from a figure in the publication) for the first and second stage, respectively. Smits et al. (1996) compared calculated CO<sub>2</sub> production based on substrate weight loss and chemical composition with measured CO<sub>2</sub> production during a solid-state fermentation using *T. reesei* QM9414 cultivated on wheat bran. The authors reported good agreement between the two values, although there was no attempt to account for other products (i.e., substrate, cells, and enzyme).

We report here on a series of bench-scale batch cellulase enzyme production experiments that were performed to test a methodology for performing carbon mass balances on cellulase production processes using both soluble sugar and insoluble cellulosic substrates. Specifically, the levels and distribution of all major carbonaceous components involved in the growth and enzyme production process (cell mass, cellulose, adsorbed and free soluble proteins, sugars, nutrients, and CO<sub>2</sub>; refer to Figure 1) were determined for a defined set of operating conditions. Our ultimate goal is to apply this methodology to facilitate kinetic model development and to guide optimization of cellulase enzyme production on pretreated lignocellulosic biomass substrates.

## Materials and Methods

**Microorganism.** A cellulose-degrading fungi, *Trichoderma reesei* (strains L27 and Rut C-30), was used in this research as previously described by Hayward et al. (1999). *Trichoderma reesei* strain L27 is a cellulase hyper-producing mutant produced by the Cetus Corporation

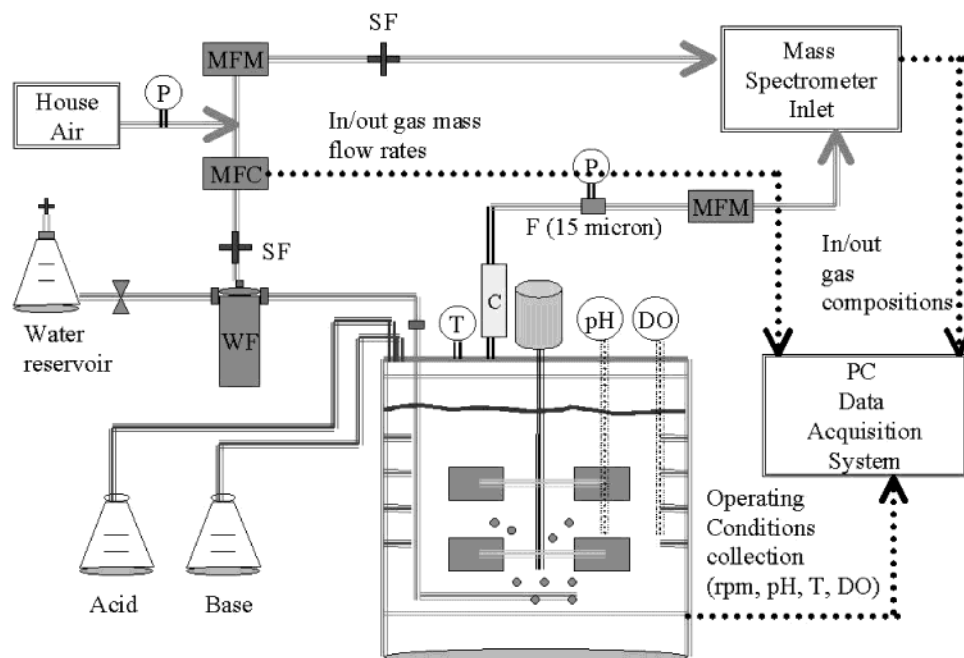
**Table 1. Growth Media Composition and Carbon Contents of Media and Cell Mass**

| component                                                                             | stage I | stage II | production fermentor | carbon contents (% w/w) |
|---------------------------------------------------------------------------------------|---------|----------|----------------------|-------------------------|
| <b>Basal Salts</b>                                                                    |         |          |                      |                         |
| KH <sub>2</sub> PO <sub>4</sub> (g/L)                                                 | 2       | 2        | 2                    |                         |
| (NH <sub>4</sub> ) <sub>2</sub> SO <sub>4</sub> (g/L)                                 | 1.4     | 1.4      | 1.4                  |                         |
| CaCl <sub>2</sub> ·2H <sub>2</sub> O (g/L)                                            | 0.4     | 0.4      | 0.4                  |                         |
| MgSO <sub>4</sub> ·7H <sub>2</sub> O (g/L)                                            | 0.3     | 0.3      | 0.3                  |                         |
| C <sub>6</sub> H <sub>5</sub> O <sub>7</sub> Na <sub>3</sub> ·2H <sub>2</sub> O (g/L) |         | 2.5      |                      |                         |
| <b>Trace Minerals</b>                                                                 |         |          |                      |                         |
| FeSO <sub>4</sub> ·7H <sub>2</sub> O (mg/L)                                           | 5       | 5        | 5                    |                         |
| MnSO <sub>4</sub> ·H <sub>2</sub> O (mg/L)                                            | 1.6     | 1.6      | 1.6                  |                         |
| ZnSO <sub>4</sub> ·7H <sub>2</sub> O (mg/L)                                           | 1.4     | 1.4      | 1.4                  |                         |
| CoCl <sub>2</sub> ·6H <sub>2</sub> O (mg/L)                                           | 3.7     | 3.7      | 3.7                  |                         |
| <b>Nitrogen Source</b>                                                                |         |          |                      |                         |
| corn steep liquor (CSL) (mL/L) <sup>a,d</sup>                                         | 10      | 10       | 10                   | 36.2                    |
| <b>Carbon Source</b>                                                                  |         |          |                      |                         |
| glucose (% w/v) <sup>b</sup>                                                          | 2       |          |                      | 40.0                    |
| Solka-floc (% w/v) <sup>c</sup>                                                       |         | 1        | 5                    | 44.6                    |
| <b>Miscellaneous</b>                                                                  |         |          |                      |                         |
| corn oil (mL/L) <sup>e</sup>                                                          |         |          | 1                    |                         |
| antifoam (mL/L) <sup>e</sup>                                                          |         |          | 0.2                  |                         |
| Tween 80 (mL/L) <sup>e</sup>                                                          |         | 1        | 1                    |                         |
| <b>Fungal Cells</b>                                                                   |         |          |                      |                         |
| mycelia (early growth stage) <sup>d</sup>                                             |         |          |                      | 50.5                    |
| spores (late growth stage) <sup>d</sup>                                               |         |          |                      | 48.0                    |

<sup>a</sup> A nutrient source for a wide variety of industrial and food fermentations and a rich source of amino nitrogen and other biologically active substances. Crude protein = 42.0% dry basis, nitrogen = 7.2% dry basis, and reducing sugars as dextrose = 5.0% dry basis. <sup>b</sup> Carbon content calculated from elemental formula. <sup>c</sup> Solka-floc solid compositional analysis (mass balance 96.28% (w/w)) yields 91.51% (w/w) cellulose and 4.77% (w/w) xylan. Carbon content calculated from elemental formulas. <sup>d</sup> Carbon content measured by CHN analyzer. <sup>e</sup> The amount of this carbon source was not significant for the overall carbon mass balance; carbon contents are not reported.

(San Francisco, CA) through classical strain improvement techniques. Strain Rut C-30 is a hyper-producing mutant developed by Rutgers (Newark, NJ).

**Culture Medium and Cultivation.** The *T. reesei* strains were grown through sporulation on V-8 plates (Schell et al., 1990). The seed cultures were grown in two stages following the methodology and medium formulations described by Mohagheghi et al. (1988). Spores from a frozen cryovial were germinated in baffled 250-mL shake flasks (50-mL working volume) containing sterile 2% (w/v) glucose nutrient medium (Table 1). After 40 h of incubation in a temperature-controlled orbital shaker (Innova 4000 Incubator Shaker, New Brunswick Scientific) operating at a temperature of 28 °C with an agitation rate of 200 rpm, the Stage I culture was transferred at a 5% (v/v) inoculum level to a Stage II baffled 1-L shake flask (300-mL working volume) containing sterile 1% (w/v) Solka-floc (commercial cellulose,



**Figure 2.** Schematic diagram of bench-scale single fermentor enzyme production experimental system: (F) exhaust gas filter; (C) condenser; (pH) pH probe; (DO) dissolved oxygen probe; (T) thermocouple; (P) pressure gauge; (WF) water filter for air saturation; (SF) sterile filter; (MFC) mass flow controller; (MFM) mass flow meter.

Fiber Sales and Development Corp., St. Louis, MO) as a carbon source (Table 1). After 48 h of incubation and the same operating conditions of Stage I, the Stage II culture was transferred at a 5% (v/v) inoculum level to a 4-L working volume production fermentor. Table 1 summarizes the media constituents for inoculum growth and enzyme production.

Cellulase production runs were conducted batchwise in a set of four bench-scale Bioflo 3000 7-L fermentors (New Brunswick Scientific, Edison, NJ) equipped with automatic temperature, agitation rate, pH, and dissolved oxygen (DO) control as well as multiple pumps for fed-batch additions. This work comprised 16 runs with *T. reesei* strain L-27, 13 using Solka-floc and 3 using glucose, and a single run with *T. reesei* strain Rut-C30 using lactose. Insoluble substrate cellulase production experiments were conducted using 5% (w/v) Solka-floc as the primary carbon source. In all runs, the temperature was kept constant at 28.0 °C, agitation was maintained between 450 and 600 rpm (controlled by agitation/DO cascade system), and the pH was maintained at 4.8 using automatic addition of alkali (4 M  $\text{NH}_4\text{OH}$ ) or acid (2 M  $\text{H}_3\text{PO}_4$ ) solutions. The level of DO was maintained at 20% or higher of air saturation using an airflow rate of 5 slpm (standard liters per minute) and agitation rates as high as 600 rpm during periods of rapid growth.

**Experimental System and Data Collection.** All Bioflo 3000 fermentors were connected to a data acquisition and control system (DACS). A PRIMA 600 mass spectrometer (Fisons Instruments, Tewkesbury, England), inlet gas mass flow controllers (MFC), and related instrumentation (MKS Instruments Inc., Andover, MA) were used to measure exhaust gas composition and estimate exhaust gas mass flow rate. Figure 2 shows a schematic diagram of the bench-scale system for a single fermentor, including the peripheral equipment employed to monitor a cellulase production run. The inlet gas supply to each fermentor is controlled using an inlet gas mass flow controller. Inlet and outlet gas stream compositions ( $\text{CO}_2$ ,  $\text{N}_2$ ,  $\text{O}_2$ , and Ar mol fractions) are obtained from the mass spectrometer. To minimize liquid volume

losses by evaporation, sterile filtered air was first sparged through a sintered metal filter into a sterilizable container (WF) filled with sterile water to saturate at room temperature the inlet air used to sparge the fermentors (Figure 2).

The use of a DACS to monitor and control enzyme production processes is advantageous because it increases the quality and quantity of experimental data that can be generated. Each fermentor communicates directly with the DACS PC, sending values from the Bioflo cabinets (agitation rate, temperature, pH, and DO) and receiving set points (airflow rate) from the computer program (Figure 2). Inlet and outlet gas stream composition data were obtained from the mass spectrometer, and the total  $\text{CO}_2$  evolved was calculated by integrating cumulative gas production over time. The inlet volumetric airflow rate ( $Q_0$ ) was measured (and controlled) using a MFC. The inlet molar flow rate ( $F_i$ ) was calculated from the volumetric airflow rate ( $Q_0$ ) reported at standard calibration conditions ( $T_s = 273 \text{ K}$  and  $P_s = 1 \text{ atm}$ ) using the ideal gas law:

$$F_i = \frac{Q_0 P_s}{RT_s} \quad (1)$$

The exhaust gas molar flow rate ( $F_0$ ) is determined as the product of the inlet gas molar flow rate ( $F_i$ ) and the ratio of nitrogen mole fractions in the inlet and outlet gas streams ( $y_{\text{N}_2,i}/y_{\text{N}_2,0}$ ):

$$F_0 = F_i \left( \frac{y_{\text{N}_2,i}}{y_{\text{N}_2,0}} \right) \quad (2)$$

The total rate of carbon dioxide evolution ( $R_{\text{CO}_2}$ ) was calculated by performing a mass balance on carbon dioxide across the fermentor:

$$R_{\text{CO}_2} = F_i \left( y_{\text{CO}_2,0} \frac{y_{\text{N}_2,i}}{y_{\text{N}_2,0}} - y_{\text{CO}_2,i} \right) \quad (3)$$

Cumulative totals (total CO<sub>2</sub>) were calculated for evolved carbon dioxide by integrating gas production over time.

Enzyme production runs were carried out for approximately 7 days, during which time samples were periodically withdrawn and analyzed to determine cellulase enzyme activity (data not reported here), total insoluble solids concentration, and the concentrations of soluble protein, glucose, and ammonia. The fraction of fungal cell mass and Solka-floc (cellulose) in the insoluble solids was determined as described below.

**Soluble Protein.** Quantification of soluble protein was carried out using the Pierce BCA protein assay kit (Pierce Chemical Company, Rockford, IL). The bicinchoninic acid (BCA) assay was conducted following the methodology and formulations described by Smith et al. (1985). Supernatants obtained by centrifugation of fermentor broth samples were analyzed for soluble protein. In addition to total soluble protein concentrations, low molecular weight (LMW) protein concentrations were also estimated by assaying samples passed through 30-kDa molecular weight cutoff filters (NANOSEP micro-concentrators). These filters retain all components with molecular weights above 30 kDa. The difference between the total soluble protein concentration and the LMW soluble protein concentration represents the concentration of high molecular weight (HMW) soluble protein. Most of the proteins comprising the cellulase complex are greater than 30 kDa in size (Coughlan, 1985), and so this HMW protein concentration provides a good estimate of the concentration of cellulase proteins.

**Soluble Sugar.** The concentrations of glucose and lactose (cellobiose and organic acid concentrations, if present) were determined from filtered sample supernatants by high-performance liquid chromatography (HP 1090 HPLC) utilizing a Biorad HPX-87H organic acids column (Biorad Labs, Hercules, CA) operating at 55 °C and a refractive index detector HP 1047A (Hewlett-Packard, Palo Alto, CA). Dilute sulfuric acid (0.1 N) at a flow rate of 0.6 mL/min was used as the mobile phase. In addition, glucose concentrations were measured using an YSI model 2700 glucose analyzer (Yellow Springs Instruments, Yellow Springs, OH).

**Total Insoluble Solids.** The concentration of total insoluble solids in culture broths was measured using a manifold filtration unit. Four milliliters of fungal whole broth samples was filtered on to preweighed Gelman glass filter (Gelman Laboratory - Pall, Ann Arbor, MI), washed twice with deionized water, and then clamped between a porous backing (to promote air flow during drying) and a glass microscope slide. This enabled the surface of the samples to remain in direct contact with the microscope slides as they dried, which has been found to produce a uniform flat sample surface that greatly improves spectroscopic scanning characteristics (Hames et al., 2002; Tholudur et al., 2002). The slides were then placed under vacuum and desiccated at room temperature until constant weight was achieved (at least 24 h). Total insoluble solids were calculated from the solids dry weight and the volume of the original sample. Duplicates of the analysis were performed at each sampling time to improve the accuracy of this measurement.

**Fungal Cell Mass and Solka-floc (Cellulose).** Fungal cell mass was estimated using a near-infrared (NIR) spectrometric technique (Hames et al., 2002; Tholudur et al., 2002) utilizing the total insoluble solids samples prepared as discussed above. Duplicate samples were scanned using the NIR spectrometer (Foss NIR Systems model 6500 forage analyzer) to obtain the cell mass dry weight percentages, which were then converted to con-

centrations by multiplying by the total insoluble solids concentration (eq 4). Solka-floc concentrations were obtained as the difference between the total insoluble solids concentration and the estimated cell mass concentration, both expressed on a dry weight basis (eq 5):

cell mass (g/L) = cell mass fraction

$$\left( \frac{\text{g cell mass}}{\text{g insoluble solids}} \right) \times \text{insoluble solids conc (g/L)} \quad (4)$$

$$\text{Solka-floc (g/L)} = \text{insoluble solids conc (g/L)} - \text{cell mass conc (g/L)} \quad (5)$$

**Adsorbed Protein.** The cellulase adsorption isotherm was first determined using Solka-floc as the cellulose adsorbent. Samples of 2.5 g of dry Solka-floc were suspended in 10 mL of 0.1 M acetate buffer, pH 4.8, and preincubated at 28.0 °C for 30 min. After preincubation, 40 mL portions of total soluble protein solution from medium filtrate, with varying protein concentrations (0.2, 0.5, 0.8, 1.1, 1.4, 1.7, 2, 4, 8, 12, 16, and 20 g/L) were added to the cellulose samples (5% [w/v] cellulose). The reaction mixture was incubated for 1 h on a rotary shaker operating at 28.0 °C and 200 rpm; this is sufficient time for equilibrium to be established (Velkovska et al., 1997). Samples were then taken from the shake flasks and centrifuged at 4800 rpm for 10 min, and supernatants were saved for soluble protein analysis. A Langmuir-type isotherm was used to fit the cellulase/cellulose adsorption data:

$$P_{\text{ads}} = \frac{P_{\text{max}} K_{\text{ad}} P_{\text{f}}}{1 + K_{\text{ad}} P_{\text{f}}} S \quad (6)$$

where  $P_{\text{ads}}$  is the protein adsorbed onto the cellulose surface (g/L),  $P_{\text{max}}$  is the maximum amount of protein adsorbed per unit weight of cellulose (g protein/g cellulose),  $K_{\text{ad}}$  is the adsorption equilibrium constant (L/g protein),  $P_{\text{f}}$  is the free protein in solution (g/L), and  $S$  is the cellulose concentration (g/L). The amount of cellulase protein adsorbed on residual cellulose was estimated from the soluble protein concentration using the best fit isotherm equation.

**Carbon Mass Balance Methodology.** Carbon mass balance closures were calculated as output carbon mass divided by input carbon mass:

$$\text{closure (\%)} = \frac{\text{carbon recovered} [\sum C_{\text{out}}] (\text{g})}{\text{initial carbon} [\sum C_{\text{in}}] (\text{g})} \times 100 \quad (7)$$

Closing the carbon mass balance for cellulase enzyme production on purified cellulose (Solka-floc) requires measuring initial and final cellulose concentrations, soluble protein concentrations, fungal cell mass concentrations, soluble sugar concentrations, cumulative gaseous carbon dioxide production (total CO<sub>2</sub>), and byproducts (organic acids, if present). Total insoluble solids, cell mass percentage in the insoluble solids (by NIR), glucose, and soluble protein were measured immediately after inoculation and thereafter every 24 h and at the end of the cultivation. In addition, the inoculum was analyzed for insoluble solids, soluble protein, and cell mass content to determine the amount of cell mass and protein actually input to the system. The carbon contribution from the medium components (primarily CSL solids, Table 1) was only included when calculating the initial input of carbon. After inoculation, it was assumed that all medium components were either consumed or converted into other

**Table 2. Average Carbon Mass Allocations for Batch Cellulase Production Runs on Soluble and Insoluble Substrates**

| run time (h)                                                                | glucose (g C) | lactose (g C) | Solka-floc (g C) | media components (g C) | fungal cell mass (g C) | soluble protein (g C) | adsorbed protein (g C) | cumulative CO <sub>2</sub> evolved (g C) | initial/recovered carbon (g) | C-balance closures (%) |
|-----------------------------------------------------------------------------|---------------|---------------|------------------|------------------------|------------------------|-----------------------|------------------------|------------------------------------------|------------------------------|------------------------|
| (A) Soluble Sugar Substrate: 5% Glucose ( $n = 3$ ) <sup>a</sup>            |               |               |                  |                        |                        |                       |                        |                                          |                              |                        |
| 0                                                                           | 82.3 ± 0.3    |               |                  | 9.8 ± 0.5              | 0.8 ± 0.1              | 0.4 ± 0.1             |                        | 0                                        | 93.5 ± 4.3                   | 100                    |
| 24                                                                          | 64.3 ± 2.4    |               |                  | 0                      | 13.0 ± 1.6             | 7.5 ± 0.2             |                        | 10.0 ± 0.8                               | 94.8 ± 12.3                  | 101.1 ± 8.3            |
| 48                                                                          | 13.8 ± 2.5    |               |                  | 0                      | 37.9 ± 2.3             | 5.2 ± 0.4             |                        | 35.7 ± 1.5                               | 92.6 ± 14.1                  | 98.8 ± 10.3            |
| 72                                                                          | 0.1 ± 0.0     |               |                  | 0                      | 40.8 ± 2.1             | 5.1 ± 0.5             |                        | 52.5 ± 2.5                               | 98.4 ± 19.2                  | 104.8 ± 15.3           |
| (B) Soluble Sugar Substrate: 5% Lactose ( $n = 1$ ) <sup>a</sup>            |               |               |                  |                        |                        |                       |                        |                                          |                              |                        |
| 0                                                                           | 0.9           | 80.3          |                  | 9.7                    | 0.7                    | 0.4                   |                        | 0                                        | 92.2                         | 100                    |
| 24                                                                          | 4.7           | 48.2          |                  | 0                      | 19.8                   | 6.4                   |                        | 11.5                                     | 90.6                         | 98.3                   |
| 48                                                                          | 0.4           | 0.0           |                  | 0                      | 40.9                   | 7.4                   |                        | 45.1                                     | 93.8                         | 101.8                  |
| 72                                                                          | 0.4           | 0.0           |                  | 0                      | 36.8                   | 7.6                   |                        | 52.9                                     | 97.7                         | 106.1                  |
| 96                                                                          | 0.4           | 0.0           |                  | 0                      | 37.4                   | 8.8                   |                        | 57.6                                     | 104.2                        | 113.1                  |
| (C) Insoluble Cellulosic Substrate: 5% Solka-floc ( $n = 13$ ) <sup>a</sup> |               |               |                  |                        |                        |                       |                        |                                          |                              |                        |
| 0                                                                           | 0.6 ± 0.3     |               | 89.1 ± 0.3       | 9.7 ± 0.1              | 0.6 ± 0.8              | 0.4 ± 0.1             | 0                      | 0                                        | 100.6 ± 0.3                  | 100                    |
| 24                                                                          | 3.1 ± 0.6     |               | 63.5 ± 9.0       | 0                      | 17.4 ± 8.5             | 3.6 ± 0.9             | 1.0 ± 0.2              | 5.9 ± 1.8                                | 94.5 ± 4.7                   | 94.0 ± 4.6             |
| 48                                                                          | 1.2 ± 1.0     |               | 20.0 ± 10.7      | 0                      | 30.5 ± 4.1             | 8.7 ± 2.4             | 0.4 ± 0.2              | 34.0 ± 6.7                               | 95.0 ± 6.3                   | 94.4 ± 6.1             |
| 72                                                                          | 0.2 ± 0.2     |               | 6.0 ± 8.8        | 0                      | 26.9 ± 5.8             | 21.1 ± 4.3            | 0.1 ± 0.1              | 56.9 ± 3.3                               | 111.2 ± 6.4                  | 110.6 ± 6.4            |
| 96                                                                          | 0.1 ± 0.0     |               | 1.2 ± 1.1        | 0                      | 22.9 ± 1.7             | 27.1 ± 3.5            | 0                      | 66.9 ± 3.4                               | 118.3 ± 9.1                  | 117.6 ± 9.0            |
| 120                                                                         | 0.0 ± 0.0     |               | 0.5 ± 0.6        | 0                      | 18.3 ± 1.0             | 29.0 ± 3.7            | 0                      | 72.3 ± 4.1                               | 120.2 ± 9.5                  | 119.5 ± 9.5            |
| 168                                                                         | 0.0 ± 0.0     |               | 0.1 ± 0.2        | 0                      | 14.5 ± 1.4             | 29.3 ± 3.3            | 0                      | 78.7 ± 4.4                               | 122.7 ± 10.2                 | 122.0 ± 10.1           |

<sup>a</sup>  $n$  = number of runs.

carbonaceous products being tracked (e.g., fungal cell mass, soluble proteins, residual sugars, or organic acids). The CSL entry in Table 1 shows the results of a CHN analysis (Perkin-Elmer CHNS/O elemental analyzer 2400, Norwalk, CT) performed to obtain the average carbon content of CSL solids and *T. reesei* L-27 cell mass in two different morphological stages (mycelium and spores). Also, a compositional analysis of Solka-floc was performed by an outside laboratory to determine cellulose and xylan content. Table 1 shows that the average cellulose and xylan levels (% dry weight) for Solka-floc correspond to an overall carbon level of 44.6% (w/w). The carbon content of the soluble proteins was estimated to be 50% (w/w) of the total protein mass (Bailey and Ollis, 1986).

Collecting and accounting for all measurements and analyses mentioned above, the C-balance closures were quantified as the ratio of total carbon recovered at any point in the cellulase production process divided by the total initial carbon measured (eq 7).

## Results

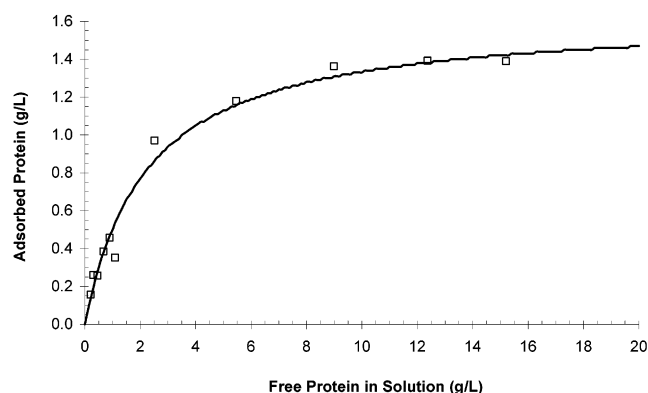
**Adsorption Characteristics of Cellulase Enzymes on Cellulose.** The adsorption characteristics of the cellulase/cellulose system were fitted to a Langmuir-type adsorption isotherm (eq 6). A nonlinear regression of the data was performed in MATLAB to estimate the adsorption parameters  $P_{\max}$  and  $K_{\text{ad}}$ . These parameters and their corresponding 95% confidence intervals were  $P_{\max} = 0.0326 \pm 0.0036$  (g protein/g cellulose) and  $K_{\text{ad}} = 0.4473 \pm 0.1553$  (L/g protein). Figure 3 shows the fit of the Langmuir-type isotherm using parameters estimated by nonlinear regression of HMW protein concentration data.

**Soluble Sugar Substrates Carbon Mass Balance.** Cultivations were performed with *T. reesei* growing on glucose or lactose to investigate carbon mass balance closures on these soluble substrates. Enzymes are not produced during growth on glucose so protein is not an important contributor to the carbon mass balance in this system. In addition, because the substrate is soluble direct gravimetric measurement of cell mass concentration is possible. The lactose substrate system is a bit more complicated because enzymes are produced; protein measurements are important but measuring cell mass concentration remains straightforward because it is a

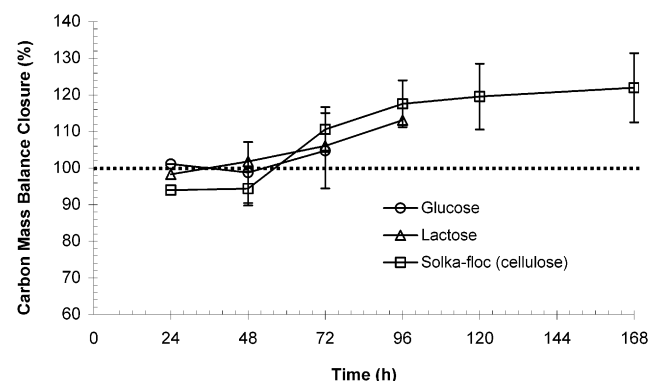
soluble substrate system. These two soluble substrate tests were conducted to validate aspects of the carbon mass balance methodology and as diagnostic tools to troubleshoot apparent measurement problems encountered late in the fermentation process.

Table 2A shows the average allocation of carbon into substrate and products along with  $\pm 1$  standard deviation during *T. reesei* growth on glucose. The table shows the distribution of all of the important carbonaceous components in the system at each sampling point. Carbon mass balances are close to 100% at each time period during the glucose fermentations (Figure 4), with the best closures,  $101.1 \pm 8.3\%$  and  $98.8 \pm 10.3\%$ , obtained at 24 and 48 h, respectively (Table 2A). The C-balance closes to  $104.8 \pm 15.3\%$  at the end of the cultivation (72 h), when the allocation of the initial carbon is  $52.5 \pm 2.5$  g in evolved CO<sub>2</sub>,  $40.8 \pm 2.1$  g in cell mass,  $5.1 \pm 0.5$  g in soluble protein, and  $0.1 \pm 0.0$  g in glucose (Table 2A). Figure 5A shows the average carbon levels and distribution of these components at each time period as a function of time. Despite reasonably accurate C-balance results throughout the process, close examination shows C-balance closures (Figure 4) trending high toward the end of the aerobic cultivation. This suggests a potential problem with one or more of the measurements.

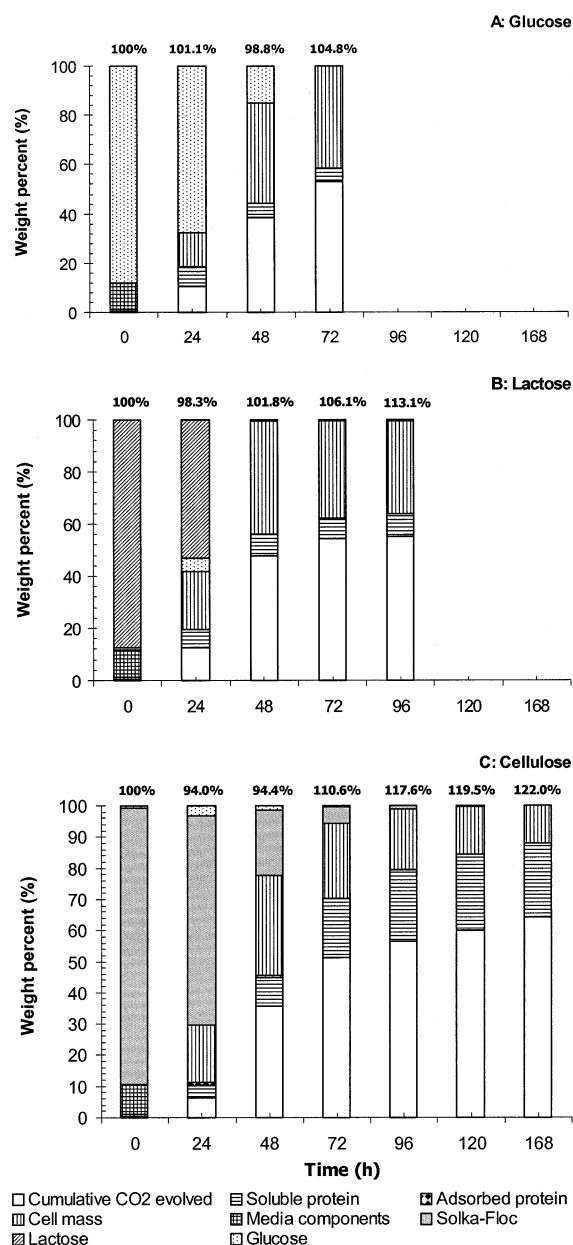
In addition to triplicate runs on glucose, a single run on 5% lactose was conducted to examine the C-balance



**Figure 3.** Adsorption isotherm of cellulase proteins on 5% Solka-floc at 28 °C. Line represents fit using adsorption parameters estimated by nonlinear regression.

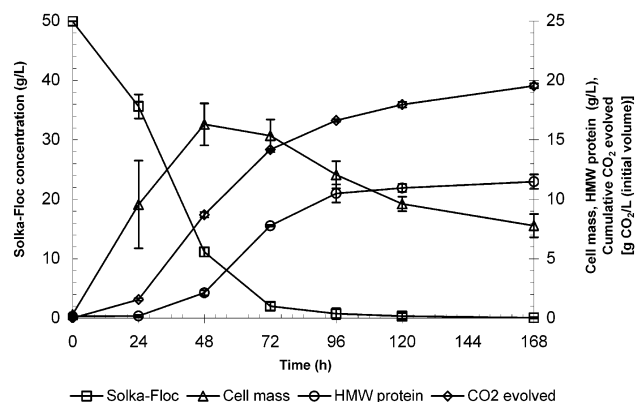


**Figure 4.** Carbon mass balance closures for *T. reesei* grown on glucose ( $n = 3$ ), lactose ( $n = 1$ ), and Solka-floc ( $n = 13$ ) as a function of run time.



**Figure 5.** Average carbon levels and distribution of all major carbonaceous components involved in cellulase production runs using both soluble sugar and insoluble cellulosic substrates: (A) 5% glucose, (B) 5% lactose, and (C) 5% Solka-floc.

for a soluble substrate system producing cellulase protein. The C-balance closure trend for this run is also



**Figure 6.** Average cellulase production kinetics on 5% Solka-floc ( $n = 13$ ).

shown in Figure 4. Although accurate C-balance closures are obtained in the lactose system at 24 and 48 h (98.3% and 101.8%, respectively), C-balance closures again trend higher later in the cultivation, reaching 106.1% and 113.1% at 72 and 96 h, respectively (Table 2B). Figure 5B shows the distribution of carbon among all major components at each time period for this run and indicates that CO<sub>2</sub> evolution is the dominant contributor to the C-balance late in the process (Table 2B). This suggests that quantitation of CO<sub>2</sub> evolution may be biased high later in the process, and this finding may also explain the trend seen in the glucose runs. Recognizing that problems may exist with the CO<sub>2</sub> production measurement at later times, the methodology was extended to evaluate the slower process of cellulase production using insoluble cellulose.

**Insoluble Cellulosic Substrate Carbon Mass Balance.** Averaged time course data for 13 replicated cellulase production runs on Solka-floc along with  $\pm 1$  standard deviation error bars are presented in Figure 6, which shows Solka-floc concentration, cell mass concentration, cellulase (HMW protein) concentration, and cumulative CO<sub>2</sub> evolution trends. The internal consistency of this batch performance data was evaluated by examining carbon mass balance closures over the course of each run. Average carbon mass allocations and carbon mass balance closures are shown as a function of run time in Figure 5C, and the distribution of each component at each time point along with  $\pm 1$  standard deviations are presented in Table 2C. As Figure 4 and Table 2C show, accurate C-balance closures were observed early in the runs, with  $94.0 \pm 4.6\%$  and  $94.4 \pm 6.1\%$  of the initial carbon accounted for at 24 and 48 h, respectively. As observed in the soluble substrate results, however, C-balance closures increase substantially to  $110.6 \pm 6.4\%$  at 72 h and then to  $117.6$ – $122.0 \pm 9.0$ – $10.1\%$  at 96 h and beyond (Figures 4 and 5C). The culture begins sporulating after 72 h (data not shown), after significant cell mass, protein, and CO<sub>2</sub> production have already occurred and the cellulosic substrate has essentially been consumed. Carbon dioxide production alone accounts for  $66.9 \pm 3.4$  g,  $72.3 \pm 4.1$  g, and  $78.7 \pm 4.4$  g of the approximate 120 g total carbon recovered at 96, 120, and 168 h, respectively (Table 2C). Thus, accurate measurement of CO<sub>2</sub> production is critical to obtaining accurate carbon mass balance closures particularly late in the process.

## Discussion

Carbon mass balancing was performed on cellulase production runs utilizing both soluble sugar and insoluble cellulose substrates. Results demonstrate the capability

to determine the levels and distribution of all major carbonaceous components at each time period during the cellulase production process using either soluble or insoluble substrates. In particular, the absolute amounts of adsorbed protein were not significant in terms of the overall carbon mass balances. As shown previously, average carbon mass balance closures were  $100 \pm 10\%$  during early stages ( $\leq 48$  h) of the cultivations using glucose, lactose, or cellulose as the substrates. Despite accurate C-balance closures observed during early stages, however, high closure values (approximately  $110 \pm 10\%$ ) were obtained at later stages of the cultivation.

As mentioned earlier,  $\text{CO}_2$  is the main carbonaceous component being produced late in the process, which is when the C-balance closures begin to substantially deviate from theoretical ( $>110\%$ ). We hypothesize that errors in quantifying  $\text{CO}_2$  are the major factor responsible for the high C-balance closures obtained late in the cultivations. Schell et al. (2002) demonstrated that errors in quantifying  $\text{CO}_2$  are more likely to influence carbon mass balances than any other source in this type of cellulase production process. Accurate measurement of  $\text{CO}_2$  production relies on suitable exhaust gas mass flow rate and composition measurements (especially  $\text{CO}_2$  and  $\text{N}_2$  mol percentages) and errors in either of these measurements could cause high  $\text{CO}_2$  production values. The mass spectrometer was calibrated periodically and check gases of known composition were run intermittently to verify instrument accuracy; the instrument was recalibrated whenever the concentration of a component in the check gas deviated from the known value by more than 0.2% (Schell et al., 2002). Thus, relative errors associated with the gas composition measurement should be small, which suggests that errors in the gas mass flow rate measurements are the major cause of problems in quantifying cumulative  $\text{CO}_2$  production. Increasing back pressure in the fermentation system as the cultivation proceeded as a result of entrained liquid or foaming plugging the gas filters may have affected gas mass flow rate measurement. Random and bias errors associated with measuring  $\text{CO}_2$  production and other parameters in this process are discussed in detail by Schell et al. (2002).

This work illustrates how applying C-balance methodology facilitates understanding process kinetics and also helps troubleshoot complex processes such as cellulase production using insoluble cellulosic substrates. The ability to accurately assess carbon mass balances over the entire course of a cellulase production process is crucial to confirm data quality, particularly if these data are to be used to support process performance guarantees or to develop and validate predictive kinetic models. Further work is needed to extend this methodology to guide optimization of cellulase enzyme production using pretreated lignocellulosic biomass as the insoluble carbon source substrate.

### Conclusions

We performed a series of bench-scale batch cellulase production experiments and developed and implemented analytical techniques to close carbon mass balances on a cellulase production process using soluble sugar (glucose and lactose) and insoluble cellulosic (Solka-floc) substrates.

Compositional analyses of *T. reesei* cells (mycelia and spores), corn steep liquor (CSL), and Solka-floc were performed in order to estimate average carbon contents. Total soluble protein and high molecular weight (HMW)

protein concentrations were quantified, and the amount of cellulase protein adsorbed on substrate cellulose was predicted using an established equilibrium adsorption isotherm. Carbon dioxide production was measured using mass spectrometry, mass flow controllers, and related instrumentation. A calibrated NIR spectroscopy-based method was used to quantify the mass fraction of *T. reesei* cells in mixed cell mass–Solka-floc solids. This ability to quantify fungal cell mass and residual cellulose levels was crucial for understanding process kinetics and facilitated calculation of carbon mass balance closures.

Accurate carbon mass balances were obtained for *T. reesei* growing on glucose or lactose during early stages ( $\leq 48$  h), with carbon closures of  $101.1 \pm 8.3\%$  and  $98.8 \pm 10.3\%$  for glucose, and  $98.3\%$  and  $101.8\%$  for lactose at 24 and 48 h, respectively. Carbon mass balance closures were acceptable in cellulase production runs on insoluble (cellulose) substrate during the early phases of the cultivation, with carbon closures of  $94.0 \pm 4.6\%$  and  $94.4 \pm 6.1\%$  obtained at 24 and 48 h, respectively. Abnormally high closures ( $>110\%$ ) were obtained toward the end of the cultivations (72–168 h), however, suggesting a measurement problem. Closer analysis suggests a suspected problem with the measurement of  $\text{CO}_2$  production, specifically, errors quantifying gas mass flow rates late in the cultivation.

Through this work, a substantially better understanding of fungal cell growth and cellulase enzyme production kinetics was accomplished. The combined set of measurements allowed carbon mass balances across the cellulase production process to be performed and the accuracy of underlying kinetic data to be assessed. Our carbon mass balance methodology and results demonstrate the capability of accounting for cellulosic solids (substrate), fungal cell mass, soluble proteins (including cellulase enzyme), adsorbed proteins (on cellulosic solids), glucose, and evolved  $\text{CO}_2$  over the course of the aerobic fermentation. This carbon mass balance methodology can be extended to facilitate a better understanding of similar biobased conversion processes utilizing lignocellulosic substrates.

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